

SIGNOS

Adaptive Input for Windows — Snapdragon AI Lab Build & Present
Challenge

QUALCOMM × HP

Control Windows with Indian Sign Language. No keyboard. No mouse.



Made with GAMMA

The Mission

What if sign language wasn't just for people-to-people communication — but a natural way to operate a computer?

Dignified Access

Good hospitality and everyday computer access for differently-abled people — built in from day one.

A New Option

SignOS complements existing assistive tools. It adds a new, dignified input layer — not a rejection of others.

Bold Vision

Sign language as a full operating system input — not a workaround, but a first-class interface.



The Gap in Existing Tools

Every current tool stops short. SignOS goes further.

Sign-to-Text

Translates signs between people. **Never touches the OS.**

Fixed Gesture Macros

One gesture, one hardcoded action — **everywhere, always.**

Chatbot Assistants

Wait for a prompt. **Don't act on your live workflow.**

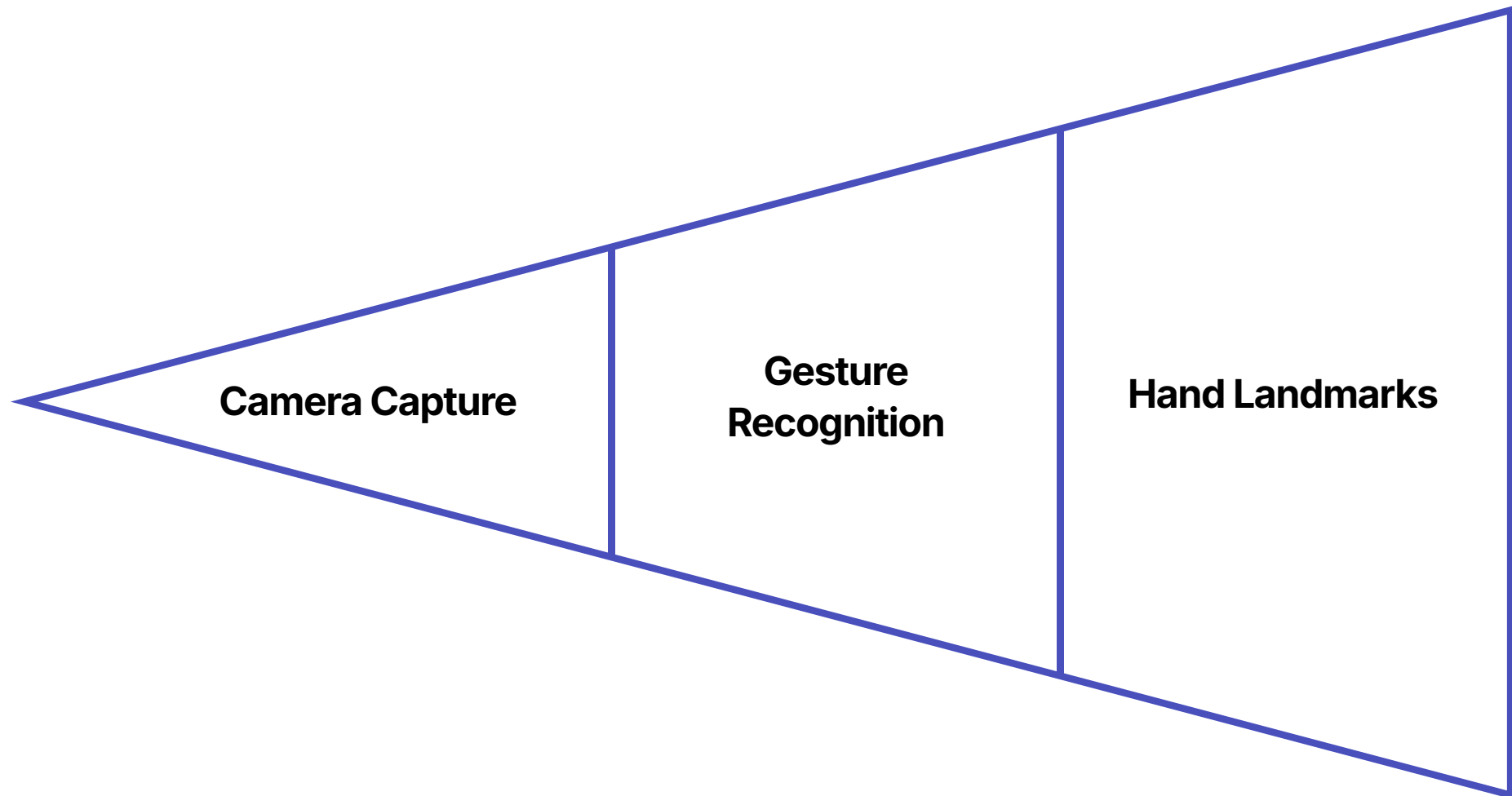
Cloud Accessibility

Needs internet. **Sends your video off-device.**

- ❏ SignOS replaces physical input entirely — and the same gesture means something different depending on which app you're in.

The Core Loop

A fully local, NPU-accelerated pipeline from camera to real Windows action.



Powered by **Snapdragon Hexagon NPU** via ONNX Runtime + QNN execution provider — entirely on-device.

Context-Aware Intent

Same gesture. Different meaning.

One vocabulary. Resolved automatically by what's on screen — deterministically, not by guessing with a large model.

Desktop

Next virtual desktop

Browser

Navigate forward

Presentation

Next slide

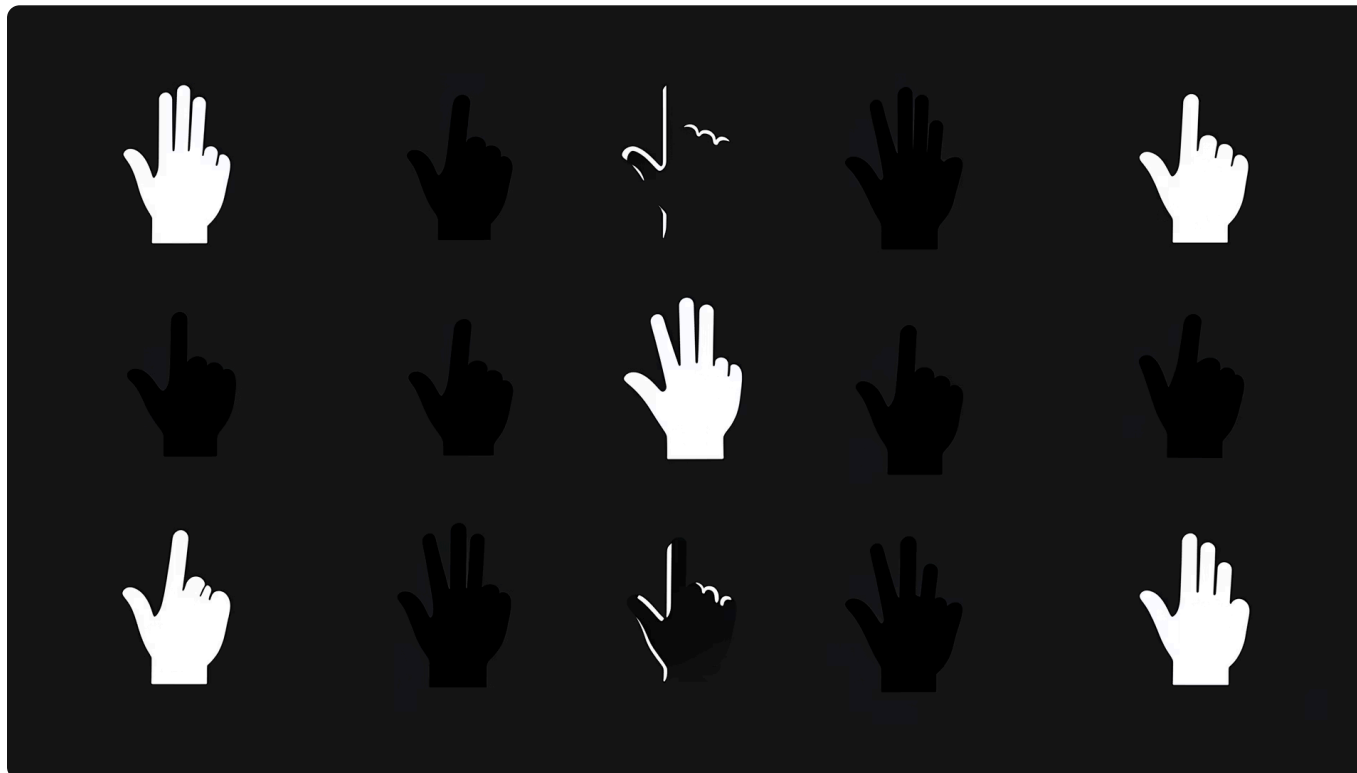
Media

Next track

↑ All triggered by a single SWIPE RIGHT gesture



Gesture Vocabulary



01

POINT

Moves the cursor

02

PINCH / PINCH+DRAG

Click or drag

03

TWO-FINGER SCROLL

Scrolls the page

04

FIST HOLD

Clutch toggle — ordinary signing never moves the cursor

05

CONFIRM GESTURE

Destructive actions always require a second, distinct gesture

Privacy & Snapdragon-Native Design



No Raw Video Stored

Only structured intent events leave the camera pipeline.



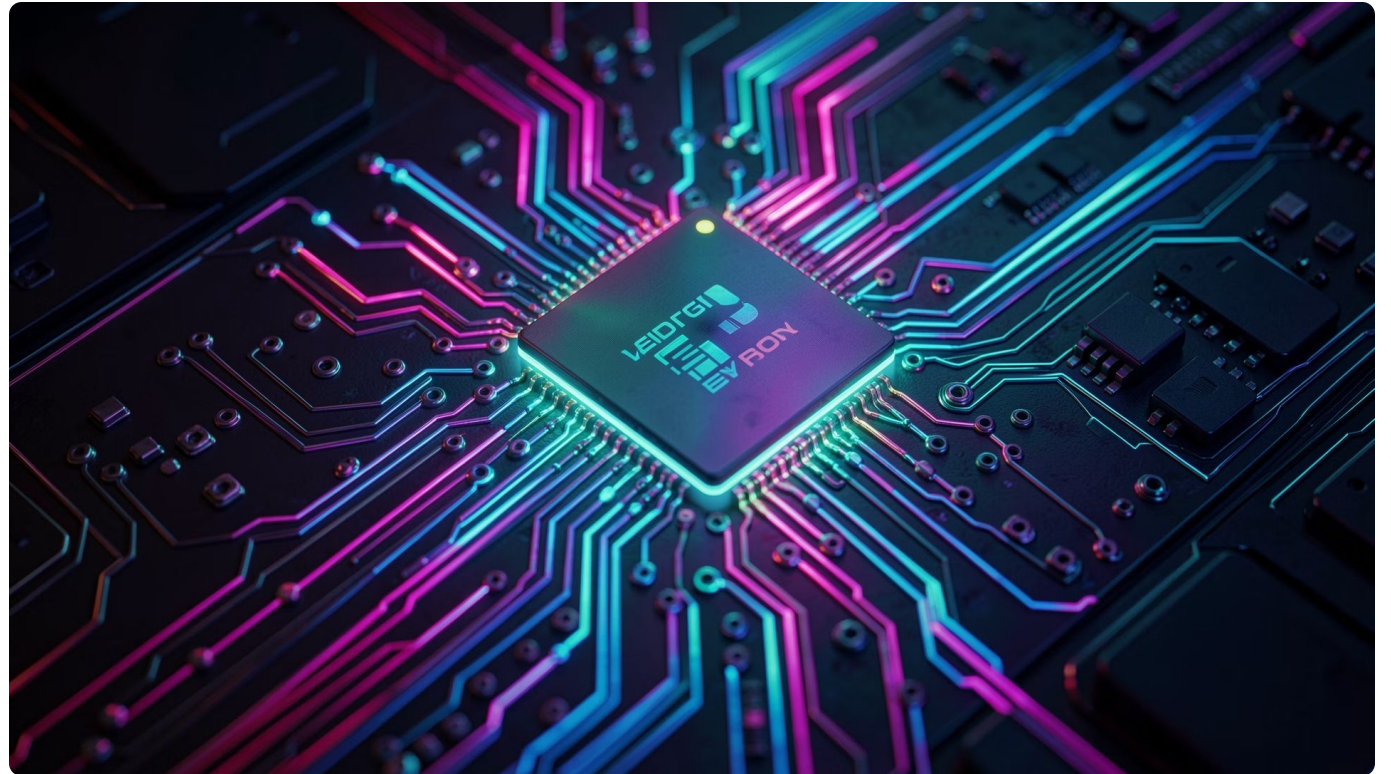
Fully Offline

Zero cloud dependency for the core control path.



Mission Control

Local dashboard shows what the system saw, inferred, and did — full transparency.



Where We Are Today

LIVE ON REAL HARDWARE

65

Automated Tests

All passing

4+

Real Windows Apps

Context-switching verified

100%

On-Device

No cloud, no latency surprise

Camera-to-gesture live. Context-aware switching confirmed. Risk-gated execution proven. Mission Control dashboard running.

Made with **GAMMA**

What's Next

ISL Vocabulary

Sourced from ISLRTC, then trained on confirmed signs

1

2

Sign Text Entry

Search, launch, and navigate apps — hands-free

NPU Profiling

Full Snapdragon NPU optimization via Qualcomm AI Hub

3

4

Routine Learning

Quietly adapts to how each person actually works

Why This Matters

A proposal for how Windows itself could work — for everyone.

SignOS isn't a gesture recognizer demo. It's a new input paradigm — built with **dignity**, **privacy**, and **real context-awareness** from day one.

Dignity

No workarounds. A first-class interface.

Privacy

Fully local. Your data stays yours.

Context

The OS understands where you are and what you mean.

